SINDH · CLASS XI · COMPUTER SCIENCE

Teach Yourself Study Guide

An in-depth companion with embedded MathJax notation and SVG diagrams for Digital Logic, SDLC, Networks, Algorithms, Python, Databases, IoT & AI, and Digital Literacy.

Read · Try · Check · Recite · Review

How to teach yourself with this book

Companion to the Sindh Textbook Board Class XI Computer Science syllabus — original explanations, not a reprint. Formulae are typeset with MathJax; circuits, stacks, and traces are inline SVG.

1. **Read for the map** (ten minutes of headings).
2. **Read for the mechanism** — cover a worked line and try it first.
3. **Try this now** boxes are mini exams.
4. **Recite** the seven gates, six SDLC phases, OSI mnemonic, three error types, four keys.
5. **Review** tomorrow, next week, next month.

> **KEY IDEA**
> Understanding is not recall. Recitation is how professionals keep protocols in their heads.

Eight-week plan

Week	Focus	Artefact
1	Unit 1 logic, gates, K-maps	LogiSim circuit + table
2	SDLC + OSI/TCP-IP	Waterfall vs Agile from memory
3	Unit 2 sorts	Two full traces
4	Search + start Python	Binary search on a register
5	Python control	Calculator + even/odd
6	Libraries + ER	ATM or report card; one ER
7	Access + IoT	Grouped query; irrigation paragraph
8	Literacy + mixed paper	Mini artefact + rehearsal exam

Unit 1 — Computer Systems

Three questions: how does a machine represent truth with two states, how do teams turn an idea into software, and how do distant machines agree on a conversation?

> **STUDY IN THREE SITTINGS**
> Logic and gates · SDLC case studies · OSI/TCP-IP. After each sitting, rebuild the main table from memory.

1.1 Discrete versus continuous

A **discrete** quantity takes separate countable values (31 students, never 31.4). A **continuous** quantity can take any value in a range (temperature, height, time). Digital electronics copies the stairs: a wire is LOW ($\mathbf{0}$) or HIGH ($\mathbf{1}$).



Figure 1.1 — Stairs (discrete / digital) versus a ramp (continuous / analog).

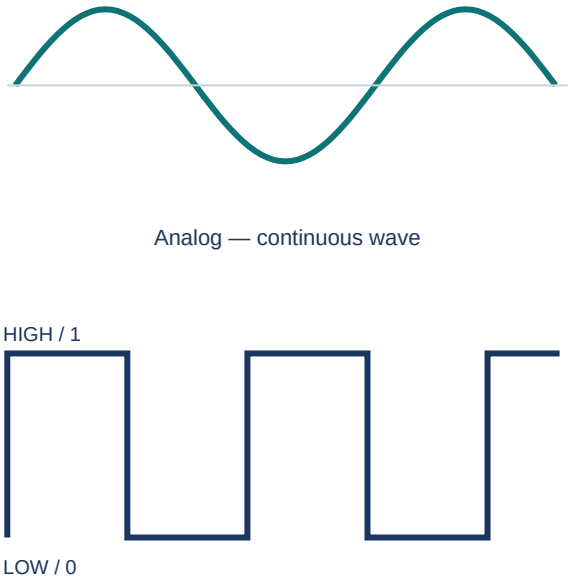


Figure 1.2 — Analog signals vary smoothly; digital signals jump between two legal levels.

Feature	Analog	Digital
Nature	Continuous, infinitely many in-between values	Discrete steps; usually two levels
Waveform	Sine-like	Square
Unit	Physical quantity	Bit: $\mathbf{0}$ or $\mathbf{1}$
Noise	Adds permanently	Ignored if it stays inside a band
Examples	Voice, vinyl, mercury thermometer	USB, HDMI, calculator

1.2 Boolean algebra

George Boole's algebra uses only $\mathbf{FALSE} = \mathbf{0}$ and $\mathbf{TRUE} = \mathbf{1}$. Precedence is

$$(\cdot) \;\rightarrow\; \mathbf{NOT}\,(') \;\rightarrow\; \mathbf{AND}\,(\cdot) \;\rightarrow\; \mathbf{OR}\,(+).$$

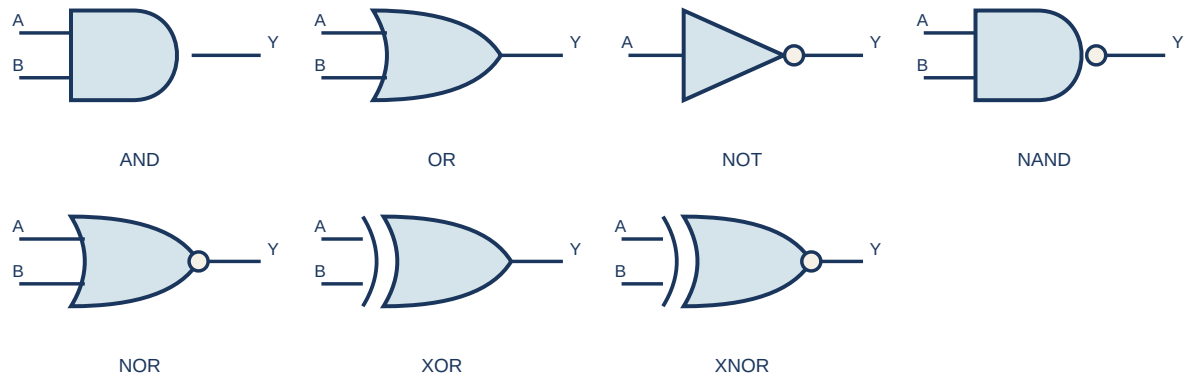
- **AND:** $Y = A \cdot B = 1$ only if every input is 1.
- **OR:** $Y = A + B = 1$ if any input is 1.
- **NOT:** $Y = A' = \overline{A}$ flips one bit.

Worked evaluations. For $A = 1,\ B = 0$: $Y = A' + B = 0 + 0 = 0$. For $A = 1,\ B = 0,\ C = 1$: $Y = (A + B) \cdot C = 1 \cdot 1 = 1$. For $A = B = C = 1$: $Y = A \cdot B + C' = 1 + 0 = 1$.

> **TRY THIS NOW**
> Evaluate $Y = (A \cdot B)' + C$ for $A = 1, B = 1, C = 0$. Answer: NAND of $1, 1$ is 0, then $0 + 0 = 0$.

1.3 The seven gates

A truth table with n inputs has 2^n rows. List inputs in binary order so no row is skipped.



Derived identities you must write in math form:

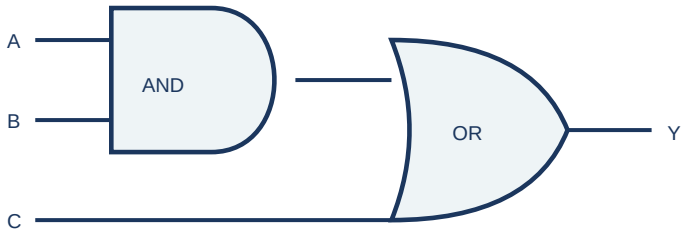
$$Y_{\text{NAND}} = (A \cdot B)', \qquad Y_{\text{NOR}} = (A + B)',$$
$$Y_{\text{XOR}} = A \oplus B = A'B + AB', \qquad Y_{\text{XNOR}} = (A \oplus B)' = AB + A'B'.$$

NAND and NOR are *universal*: any Boolean function can be built from only NAND, or only NOR.

A	B	AND	NAND	OR	NOR	XOR	XNOR
0	0	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	0	1	0	0	1

1.4 Expressions, diagrams, SOP

For $Y = A \cdot B + C$ (three variables $\Rightarrow 8$ rows) compute the helper column $A \cdot B$, then OR with C.



Y = (A · B) + C

Figure 1.3 — AND has higher precedence, so it is drawn first; its output feeds OR together with C.

Sum of products: each row where $Y = 1$ is a minterm. If a variable is 0 in that row, complement it. Then

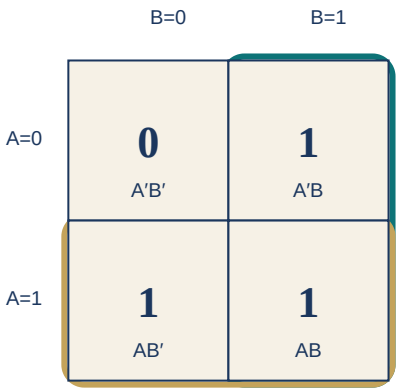
$$Y = \sum \text{minterms of the 1-rows.}$$

Example: 1-rows $AB = 01$ and 11 give $Y = A'B + AB = B$.

> **COMMON MISTAKE**
> In a minterm, a 0 in the row means the variable is complemented. Row $A = 0, B = 1$ is $A'B$, not AB'.

1.5 Karnaugh maps

Adjacent cells (including wrap-around) differ by one bit — **Gray code**. Group 2^n ones; the changing variables cancel.



Groups: right column → B, bottom row → A, so Y = A + B

Figure 1.4 — Two-variable map for Y = A′B + AB′ + AB.

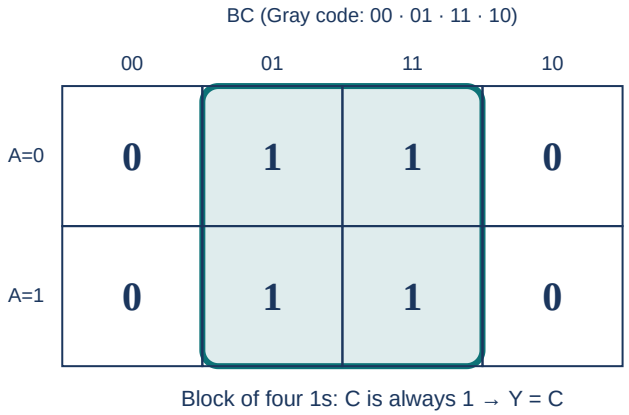


Figure 1.5 — Three-variable map. Columns are BC = 00, 01, 11, 10 — never 00, 01, 10, 11.

TRY THIS NOW
Simplify $Y = \sum m(0,1,2,3,5,7)$. Two groups of four give $Y = A' + C$.

1.6 Software Development Life Cycle

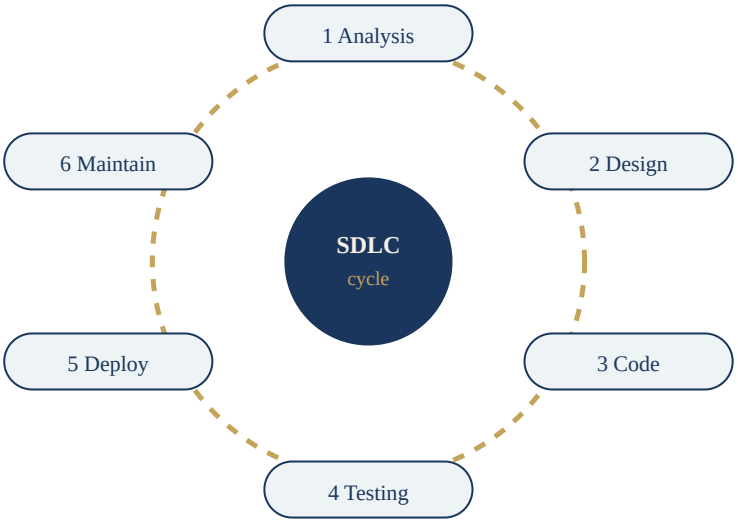


Figure 1.6 — The six SDLC phases form a maintenance cycle, not a dead end.

Phase	Activity	Deliverable
1 Analysis	Meet users; list features	SRS
2 Design	HLD/LLD, screens, ER, UML	Design document
3 Implementation	Code, Git, comments	Working build
4 Testing	Unit, integration, system, UAT	Test report
5 Deployment	Direct, phased, pilot, parallel	Live software
6 Maintenance	Corrective, adaptive, perfective, preventive	Patches

CAPP maintenance: Corrective (login bug), Adaptive (new Windows), Perfective (dark mode), Preventive (daily backup).

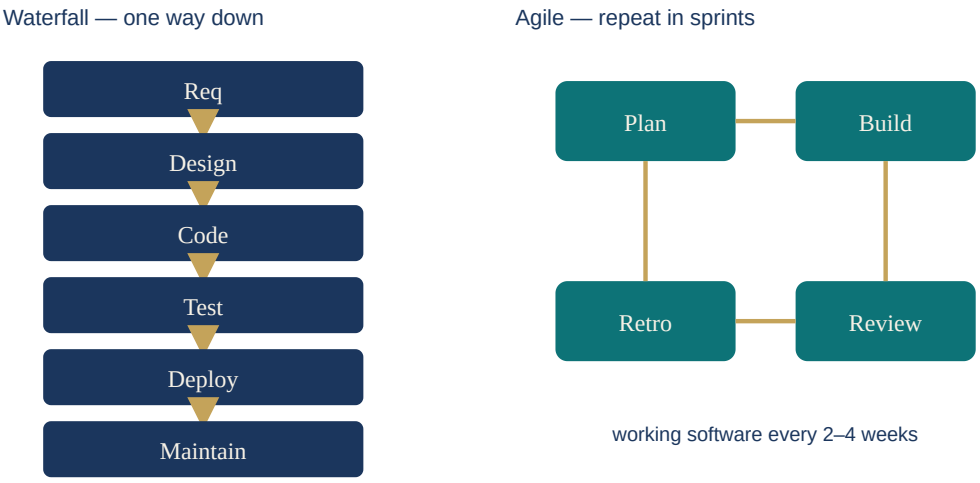


Figure 1.7 — Waterfall finishes a phase before the next begins. Agile repeats a short sprint.

Al-Noor Library (fixed 2,400-book system, offline Windows, six months) fitted **Waterfall**. The student mobile app (unclear features, five two-week sprints, Flutter) fitted **Agile / Scrum**. Same school, two models — match the model to the project.

1.7 Communication models

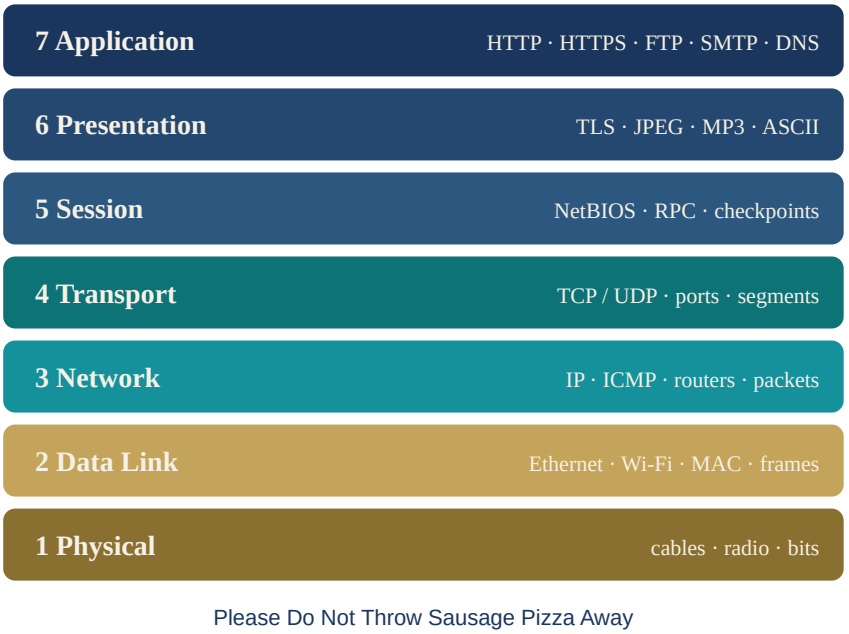
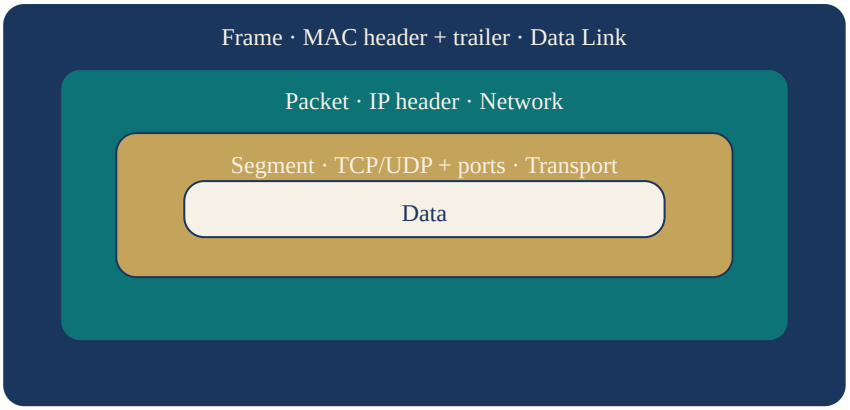


Figure 1.8 — OSI seven-layer stack with typical protocols.



Going down: wrap. Going up: unwrap.

Figure 1.9 — Encapsulation names: data → segment → packet → frame → bits.

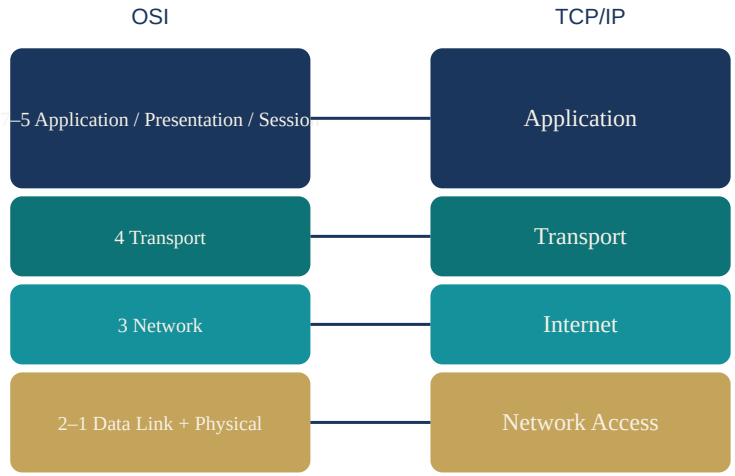


Figure 1.10 — TCP/IP collapses OSI 7–5 into Application and OSI 2–1 into Network Access.

TCP is connection-oriented and reliable (web, email, files). UDP is connectionless and cheap (live video, games). An **IP address** is logical; a **MAC address** is burned into the NIC. Routers live at Layer 3; switches at Layer 2; hubs at Layer 1. `ping` uses ICMP at the Network layer.

Unit 2 — Computational Thinking and Algorithms

Computational thinking is thinking so clearly that a machine — or another human — can follow you.

An **algorithm** is a finite, language-independent sequence of steps. **Pseudocode** writes those steps with IF/FOR/WHILE but no compiler syntax.

- **Decomposition** — split “learn Spanish” into alphabet, vocabulary, grammar, practice.
- **Pattern recognition** — a triangle of stars that grows by one star per row is a loop $i = 1 \ldots n$.
- **Abstraction** — making tea is boil, add, steep, pour. Ignore mug colour.

> **EXAM TIP**
> Ignoring the kettle brand is abstraction, not decomposition.

2.1 Bubble Sort

Compare neighbours; swap if out of order. After pass k, the k largest items sit at the right. Nested loops give time that grows like n^2.



Figure 2.1 — Trace of {8, 4, 1, 9, 3} in ascending order. Teal cells are already parked.

```
FOR pass ← 0 TO n−2
    FOR i ← 0 TO n−2−pass
        IF DATA[i] > DATA[i+1] THEN swap
```

Strengths: in-place and stable. **Weaknesses:** time grows like n^2; many swaps.

2.2 Selection Sort

On pass k, find the minimum of the unsorted tail and swap it into index k. At most one write per pass — fewer swaps than Bubble Sort, still $\Theta(n^2)$ comparisons, not stable.

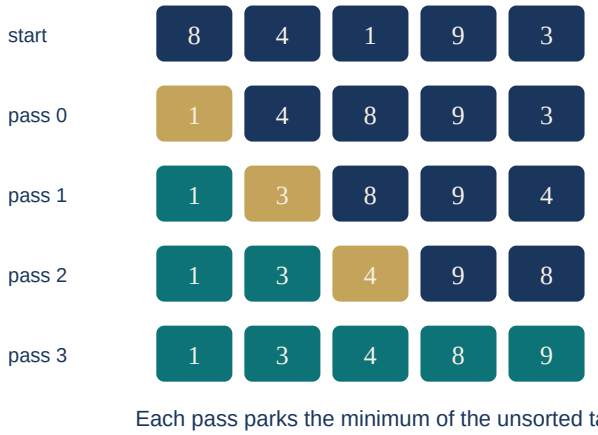


Figure 2.1b — Same list under Selection Sort. Gold is the newly parked minimum.

2.3 Searching

Linear search checks items $1, 2, \ldots, n$. Works on unsorted data. Worst case n probes.

Binary search requires a sorted array. Each step uses

$$\text{mid} = \left\lfloor \frac{\text{low} + \text{high}}{2} \right\rfloor.$$

If the target is smaller, set $\text{high} = \text{mid} - 1$; if larger, $\text{low} = \text{mid} + 1$. About $\log_2 n$ probes — twenty looks for a million items.

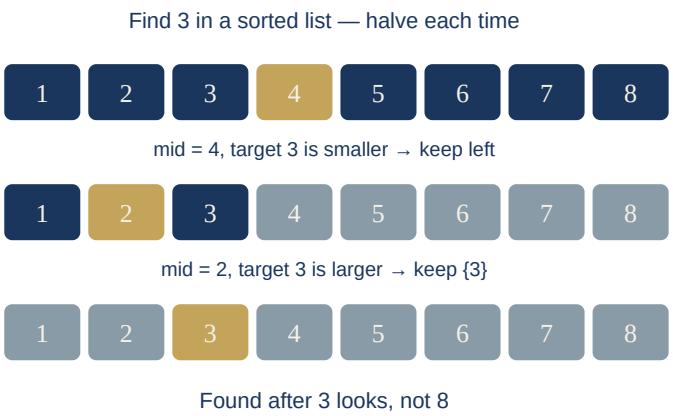


Figure 2.2 — Binary search for 3. Gold cell is mid; grey cells have been discarded.

> **COMMON MISTAKE**
> Binary search on an unsorted list is not "a bit worse." It is wrong: the discarded half may hold the target.

	Linear	Binary
Needs sorted data?	No	Yes
Idea	One-by-one	Halve the interval
Best for	Small or unsorted; one-off search	Large static sorted lists
Growth	$O(n)$	$O(\log n)$

Judge every algorithm on **correctness** (right for every valid input, and it finishes), **efficiency** (time and memory as n grows), and **clarity**.

Unit 3 — Programming Fundamentals

A computer does exactly what you said, including the mistakes. Python is the high-level language this course uses because it reads almost like English.

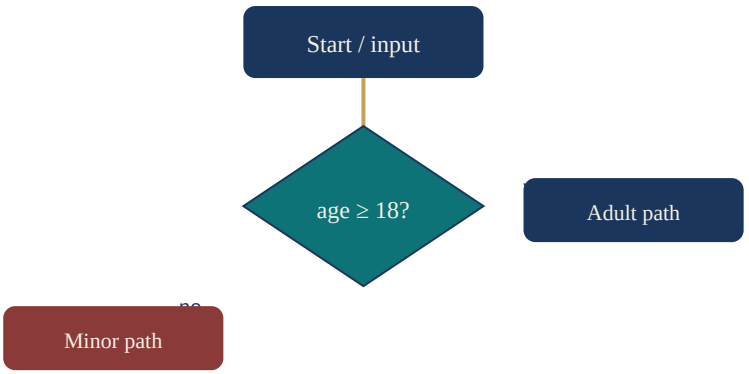


Figure 3.1 — Selection: exactly one path runs.

`input()` always returns a **string**. Convert with `int()` or `float()`. Indentation is syntax: 4 spaces per block.

Family	Symbols	Notes
Arithmetic	+ − * / // % **	$9//2 = 4$, $9\%2 = 1$, $2^3 = 8$
Assignment	= += −= *=	After a = 10; a += 5, a is 15
Relational	== != > < >= <=	Return True/False
Logical	and · or · not	Combine conditions
Membership	in · not in	"P" in "Python" is True
Bitwise	& | ^ ~ << >>	Same truth tables as Unit 1 gates

Bitwise worked example. $a = 10 = 1010_2$, $b = 6 = 0110_2$:

$$a \wedge b = 0010_2 = 2, \quad a \vee b = 1110_2 = 14, \quad a \oplus b = 1100_2 = 12, \quad a \ll 1 = 20 = a \cdot 2.$$

Also $\sim x = -x - 1$ in two's complement, so $\sim 10 = -11$.

> **COMMON MISTAKE**
> = stores a value. == tests equality. Using input() in arithmetic without int()/float() is the classic first-year crash.

3.1 Control and libraries

for when the trip count is known (`range(start, stop, step)`). **while** until a condition dies. `break` leaves now; `continue` skips this iteration.

```
import math, random, datetime
print(math.sqrt(25), math.pi)
secret = random.randint(1, 10)
now = datetime.datetime.now().strftime("%d-%m-%Y")
```

Third-party libraries need `pip install` (NumPy, Pandas, Matplotlib).

Error	When	Example
Syntax	Will not start	Missing), bad indent, if without :
Runtime	Dies while running	10/0, int("abc")
Logical	Runs and lies	i -= 1 in a loop that should climb — infinite loop

Read a traceback from the bottom: last line is the disease.

Unit 4 — Data and Analysis

Eighty, seventy-five, ninety is data. "The class average is 80%" is information. A database is the organised cupboard that keeps those facts consistent.

A **DBMS** (Access, MySQL, Oracle, SQL Server) inserts, updates, deletes, and retrieves with less redundancy, better consistency, security, and integrity. An **RDBMS** stores related tables.

Key	Role
Candidate	Any unique non-null identifier — a table may have several
Primary	The chosen candidate; exactly one; never NULL
Alternate	A leftover candidate
Foreign	Points at another table's primary key; may repeat

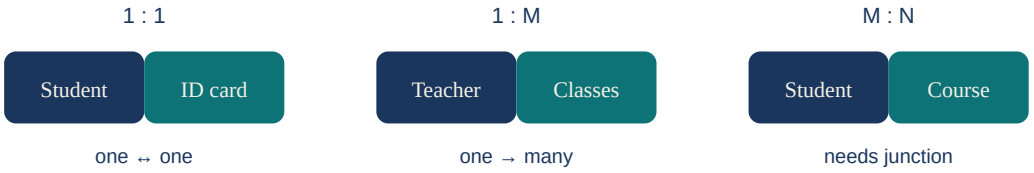


Figure 4.1 — Three cardinalities. Only M:N needs a junction table.

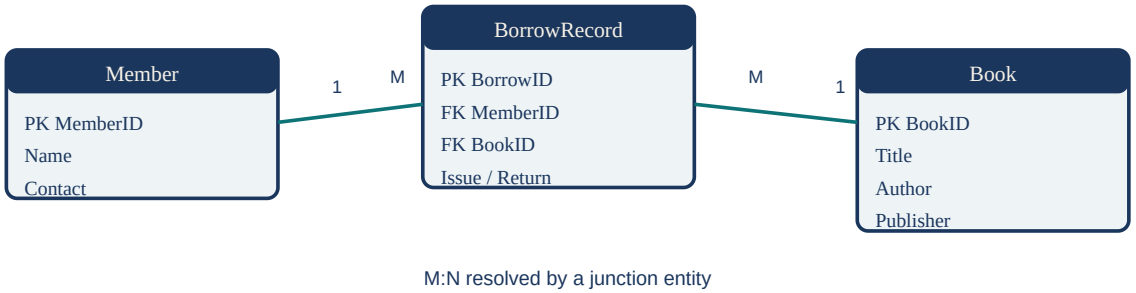


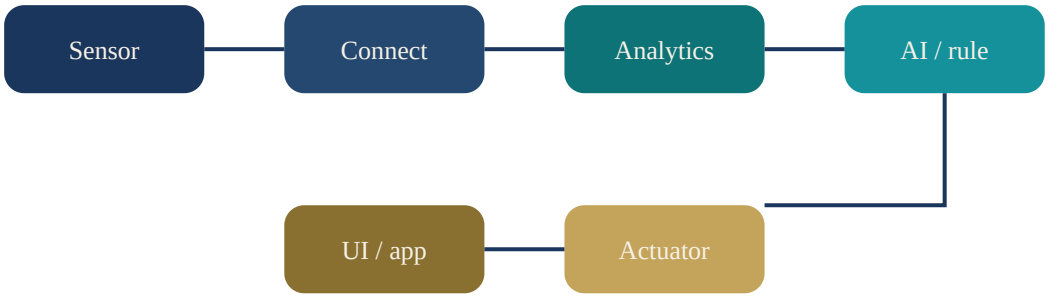
Figure 4.2 — Library ER: Member and Book are M:N, resolved as BorrowRecord.

Referential integrity: every foreign key must match a living parent. *Cascade update* rewrites child FKs when the parent PK changes. *Cascade delete* removes child rows when the parent is deleted — use it like a chainsaw.

Access objects: **table** stores, **form** enters, **query** asks, **report** prints. In Query Design, AND is the same criteria row; OR is different rows. Totals Σ: Group By + Count/Avg/Sum/Min/Max. Line charts show trends; pies show parts of a whole; bars compare named categories.

Unit 5 — Applications and Impacts of Computing

IoT collects, analytics explains, AI decides. Treat those three words as a loop, not as synonyms.



Collect → carry → explain → decide → act → show

Figure 5.1 — Smart irrigation: moisture sensor → radio → cloud analysis → valve actuator → farmer's phone.



Raw marks become a pass/fail chart only after cleaning

Figure 5.2 — Analytics turns raw records into a decision only after cleaning.

	IoT	Analytics	AI
Job	Sense and actuate	Find patterns	Decide or predict
Pieces	Sensors, actuators, links, cloud, UI	Collect–store–clean–analyse–visualise	Models, learning
Example	Soil probe, smartwatch	Pass/fail pie chart	Face unlock, recommendation

Sensor ≠ actuator. A moisture probe does not open the tap; the valve does. Connectivity (Wi-Fi, Bluetooth, 4G/5G, LoRa) carries the message. Cloud stores the history. The UI is how a human looks and commands.

5.1 Sources and impacts

Kind	Meaning	Example
Primary	First-hand original	Interview, NADRA CNIC capture
Secondary	Already explained	Textbook, PBS census summary
Tertiary	Quick digest	Encyclopedia, many Wikipedia leads

A nameless social-media image with no date is simply unreliable. Computing helps (recorded lessons, telemedicine, digital banks) and harms (screen fatigue, leaked records, lost jobs). Responsible use: strong passwords, no rumour-forwarding, planned offline time.

Assistive tech — mobility (wheelchair), sensory (NVDA screen reader), communication (picture boards), cognitive (audiobooks). They are how computing keeps its promise to everyone.

Unit 6 — Digital Literacy

Digital literacy is not "I can open Chrome." It is find, judge, collect, and show — without stealing words or leaking private data.



Figure 6.1 — Digital inquiry workflow from a sharp question to a final artefact.

	Qualitative	Quantitative
Nature	Words, reasons, feelings	Numbers you can count
Ask	Why / how	How many / how much
Example	"Notifications break my focus"	4 hours/day, 72% distracted

Primary data is what you gather for this question. **Secondary data** already exists (a PBS table). Advanced search: quotes for a phrase, `filetype:pdf` for reports, `site:.edu` or `site:.gov.pk` for a domain family.

1. One sharp research question.
2. Qualitative, quantitative, or mixed.
3. Method that can actually answer it (interview, survey, observation, prototype, simulation).
4. A sample that resembles the group.
5. Tools: form, script, sheet.
6. Collect without steering people toward your favourite answer.
7. Store in a spreadsheet; keep a backup.
8. Ethics: permission, no extra personal fields, no silent copying (plagiarism).

A decorated Canva file is not an artefact if it never answers the question. Evidence → claim → recommendation.

Case studies and closed-book drills

Al-Noor Library — why Waterfall worked

1,200 students, 2,400 books, paper register, lost copies, handwritten overdue lists. Requirements were frozen: add/edit books, search, 14-day loans, overdue alerts, printed reports, offline Windows. Team of four, six months, Python + Tkinter + SQLite. Waterfall fitted because nothing experimental was

asked. Testing (black box) found a search crash on misspellings and an overdue alert that ran only if a button was clicked. Both were fixed and retested. A two-week **pilot** sat on the librarian's desk before students used it. Later maintenance: printer-driver fix (corrective), e-book category (adaptive), faster search (perfective), daily USB backup (preventive). Overdue returns fell about 70%.

Al-Noor Student App — why Agile worked

Nobody could list every feature on day one. Five two-week sprints, Flutter, five-person Scrum team. Sprint 1 shipped login + timetable; feedback added teacher names. Sprint 2 shipped exam results; parents then asked for a full attendance record. Sprint 3 shipped SMS absence alerts; when the gateway died for two days the team switched to email and kept moving. Sprint 4 connected to the library through an API and added an available-copies counter in thirty minutes. Sprint 5 shipped homework photos and fixed three launch bugs. Five of ten original stories shipped in ten weeks. That flexibility is the point of Agile.

> **DECISION RULE**
> Pacemaker firmware with a frozen regulator spec → Waterfall (late change can kill). A club-events app that will grow with student requests → Agile. Four lines in a notebook is a complete 4-mark compare.

Logic studio — $Y = (A + B)' \cdot C$

NOR of A, B feeding AND with C. Eight rows; $Y = 1$ only on $A = B = 0,\ C = 1$. One minterm: $A'B'C$.

A	B	C	A+B	(A+B)′	Y
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

K-map wrap-around: $Y = \sum m(0, 2, 8, 10)$ occupies the four corners. Shared literals are B' and C'; A cancels. Result $Y = B'C'$. Forget wrap-around and you leave four lonely minterms.

Protocol flash cards

If you see…	Layer	Why
RJ45, Cat6, voltage, fibre pulse	Physical	Raw medium
MAC, switch, Ethernet frame, CRC	Data Link	One LAN hop
192.168.1.1, ping, router, OSPF	Network	Path across nets
Port 443, handshake, segment	Transport	Which app, reliability
Login session, RPC	Session	Conversation life
TLS, JPEG, MP3, UTF-8	Presentation	Format / secret / size
Browser, DNS lookup, SMTP send	Application	What the user asked

Algorithm studio — same list, two sorts

Bubble Sort on $\{6, 2, 5, 1\}$: pass 1 → $\{2, 5, 1, 6\}$ (three swaps); pass 2 → $\{2, 1, 5, 6\}$; pass 3 → $\{1, 2, 5, 6\}$. Selection Sort finds min 1 and swaps once with 6 → $\{1, 2, 5, 6\}$. Fewer writes. Watch equal keys: Bubble Sort is stable; Selection Sort may swap two 4s.

Binary search for 11 in $\{2, 5, 7, 11, 13, 17, 19\}$: $\mathbf{mid} = \lfloor (0 + 6)/2 \rfloor = 3$, hit on the first probe. For 2 (left edge) you need three looks. For 12 (absent) you finish with $\mathbf{low} > \mathbf{high}$ and report not found — write that failure case so you do not invent a crash.

Move	Strategy
Split “build the app” into login, marks, notices	Decomposition
Every notice has a title, date, body — same shape	Pattern recognition
Ignore button colour when specifying “post a notice”	Abstraction
Write numbered steps before opening VS Code	Algorithm

Python studio — ATM skeleton

```
PIN = "1234"
balance = 10000.0
tries = 3
ok = False
while tries > 0 and not ok:
    entered = input("Enter PIN: ")
    if entered == PIN:
        ok = True
    else:
        tries -= 1
        print("Wrong PIN!", tries, "left.")
if not ok:
    print("Card locked.")
else:
    running = True
    while running:
        print("1 Check  2 Deposit  3 Withdraw  4 Exit")
        choice = input("Choose: ")
        if choice == "1":
            print(f"Balance: Rs. {balance:.2f}")
        elif choice == "2":
            amt = float(input("Deposit: "))
            if amt > 0:
                balance += amt
        elif choice == "3":
            amt = float(input("Withdraw: "))
            if 0 < amt <= balance:
                balance -= amt
            else:
                print("Rejected.")
        elif choice == "4":
            running = False
```

Sequence + selection + repetition + bool flags. Nested-if cousin: a guarded calculator that refuses $/0$. Infinite-loop cousin: `i = 1; while i < 6: print(i); i -= 1` — a logical error, legal Python that never dies.

Clinic ER (15 minutes on paper)

Patient and Doctor are M:N, resolved as Appointment(AppointmentID PK, PatientID FK, DoctorID FK, VisitDate, Fee). Cascade-delete of a doctor wipes history — forbid the delete or mark inactive. Customer 1—M Order needs no junction: one order belongs to one customer. AND is the same criteria row; OR is different rows.

Need	Access Design grid
CS students only	Department Criteria: "Computer Science"
IT or CS	Criteria / or: two department values
Aslam in IT	Name and Department on the SAME row
Headcount per dept	Totals on; Department Group By; StudentID Count
Average marks per dept	Department Group By; Marks Avg

Society and literacy studios

Memorise the smart-home skeleton: sensor → Wi-Fi → hub/cloud → analytics vs set-point → rule/AI → actuator → phone UI. If Wi-Fi dies the chain breaks — that one failure sentence shows you understand dependency.

Exhibit	Verdict
Your interview with the librarian	Primary source / primary data
PBS graph of literacy rates	Secondary
Wikipedia lead paragraph	Tertiary; start only
Anonymous “forward this” image	Unreliable — no author or date
Government .gov.pk PDF, 2024	Usually reliable secondary

Impact formula: field + one help + one harm + one responsible action. Mini inquiry (90 minutes): one question, two advanced searches, eight mixed items, 15 classmates with permission, one chart, one recommendation aimed at a real audience.

Unit self-tests (closed book)

Unit 1. How many legal levels has a digital signal? Write $A \oplus B$ as a sum of products. Why is NAND universal? First SDLC deliverable? Which OSI layer adds ports / IP / MAC? Why might a cricket stream use UDP?

Unit 2. Define an algorithm. Which CT strategy ignores mug colour? Why does Selection Sort often swap less than Bubble Sort? One limitation of binary search. Name correctness, efficiency, clarity.

Unit 3. Legal comment syntax? Why is `my-var` illegal? What does `input()` return? Standard indent? Loop when the count is known? Bitwise AND? Three error types?

Unit 4. DBMS besides Access? ER rectangle? Relationship that needs a junction? Object for retrieval vs friendly entry? Aggregate for a mean? Define a foreign key. Cascade update? Chart for a monthly trend?

Unit 5. Watch → phone heart-rate is mainly which tech? Device that judges dry soil vs opens the valve? Duplicate rows are which analytics step? Interview is which source? Two reasons a website is unreliable? One help and one harm in healthcare? What does a screen reader do? Six IoT components?

Unit 6. Digital literacy in one sentence. “Notifications wreck my focus” — qualitative or quantitative? Best PDF search for internet use in Pakistan? Your own survey replies? What does `site:` do? Copying without credit? Why ask permission?

Selected answers

Digital: two levels, a noisy 1 is restored. $A \oplus B = A'B + AB'$. NAND builds any gate. SRS. Transport / Network / Data Link. UDP: late frames are useless. Algorithm: finite language-free steps. Abstraction. Selection: at most one swap per pass. Binary search needs sorted data. input → str. 4 spaces. for. &. Syntax, runtime, logical. MySQL/Oracle/SQL Server. Entity. M:N. Query / Form. Avg(). FK points at another PK. Child FKs follow a parent PK change. Line chart. IoT. Sensor /

valve. Cleaning. Primary. No author or date. Telemedicine / leaked files. Speaks on-screen text. Sensors, actuators, connectivity, processing, cloud, UI. Quotes + filetype:pdf. Primary data. Restrict domain. Plagiarism. Consent.

Studios, rehearsal paper, and recitation cards

Burglar alarm

Armed switch A, door D, window W. Siren

$$Y = A \cdot (D + W).$$

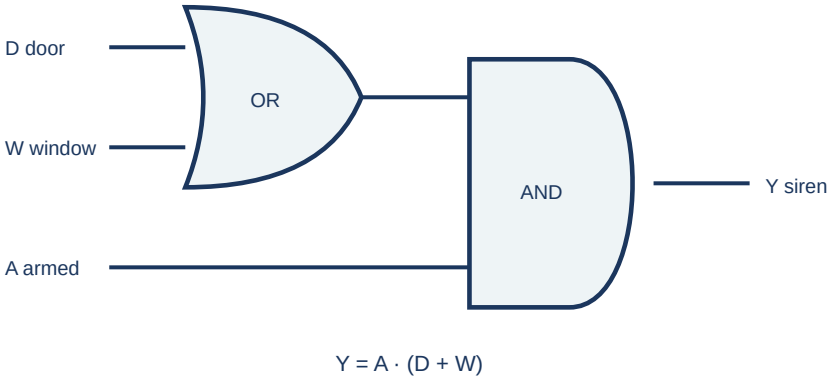


Figure S.1 — OR of the contacts, then AND with the armed switch.

If $A = 0$ the house stays quiet. Build it in LogiSim and poke all eight rows.

Final week — mixed drills without a teacher

Day 1 (logic + network). Write all seven 2-input tables from a blank page. K-map $Y = \sum m(1, 3, 5, 7)$ should collapse toward C. List OSI layers both ways with one protocol each. Explain HTTPS encapsulation in five sentences.

Day 2 (algorithms + Python). Trace both sorts on $\{8, 3, 6, 2\}$. Binary-search 6 in $\{1..9\}$ writing every mid. Code a calculator and a guess game; break them on purpose; read the traceback. Predict: $9//2 == 4$, $9\%2 == 1$, `"Py" in "Python"`, `not (True and False)`.

Day 3 (data + society + literacy). Draw library, clinic, or bus ER. Write AND vs OR query rows. Irrigation long answer in six labelled sentences. Classify five sources on your phone. Sit the 40-mark paper again. Target 32.

40-mark rehearsal (60 minutes, closed book)

A (10×1). Digital levels? XOR 1 when? First SDLC phase? TCP/IP layer count? Search that needs sorted data? `input()` type? ER rectangle? Soil dryness device? Your own survey replies? `site:` does what?

B (6×2). Analog vs digital · when Waterfall · why binary search fails on {9, 1, 4, 7} · = vs == · referential integrity · NADRA interview vs textbook paragraph.

C (3×6). Truth table of $Y = A' \cdot B + C$ plus SOP · Bubble-sort trace of {6, 2, 5, 1} · Sindh irrigation chain naming sensor, link, analytics, AI, actuator, UI.

Scheme A: two values; different; analysis; 4; binary; str; entity; sensor; primary; restrict domain.

Recitation cards

A Gates. AND all 1s · OR any 1 · NOT flip · NAND not-all-1s · NOR only all 0s · XOR different · XNOR same · rows 2^n.

B SDLC. Analyse → Design → Code → Test → Deploy → Maintain (CAPP).

C OSI. Please Do Not Throw Sausage Pizza Away. ping = ICMP = layer 3.

D Algorithms. Bubble neighbours · Selection parks min · Linear any list · Binary sorted only · $\text{mid} = \lfloor (lo + hi)/2 \rfloor$.

E Python. input → str · indent 4 · syntax / runtime / logic.

F Data. PK unique · FK points home · M:N + junction · AND same row · OR different rows.

G Society. Collect / explain / decide · sensor ≠ actuator · consent · no plagiarism.

The board paper rewards people who can do a truth table, trace a sort, write a small program, draw an ER fragment, and judge a source. If you have worked the amber boxes twice, you are the teacher you were waiting for.